

Vinh-Hung (Tommy) Ly

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Education

Columbia University, New York

May 2025

B.A. Computer Science, Minor in Statistics

GPA: 3.7/4.0

Coursework: Probabilistic Machine Learning (PhD), Statistical Analysis of Neural Data (PhD), Bayesian Statistics, Algorithms, Stochastic Processes, Time Series, Artificial Intelligence

Honors: Knight-Hennessy Scholars semifinalist (top 6% of 8,570 applicants); Cohen Scholar (\$25k); Columbia GS Honor Society (top 76); Dean's List (6 semesters)

RMIT University Vietnam – Bachelor of Commerce

2013

Industry Experience

Evolve, New York, NY

Jun 2025 - Present

Data Engineer

- Cut Snowflake compute cost on four high-traffic dbt models by converting table → incremental materializations, achieving up to 49x speedup (740s to 15s); tuned 11 additional queries across revenue and forecasting repos.
- Migrated the production pricing pipeline onto a new demand-forecasting data feed over 25+ shipped changes, with shadow-to-prod parity testing, deduplication, and guardrails.
- Built the scheduling and email-digest layer for an internal FastAPI app: replaced page-load triggers with an in-process APScheduler job; built a headless screenshot to email pipeline; debugged a refresher-lock deadlock.

Uber Technologies, New York / Singapore / Vietnam

2016 - 2020

Senior Regional Operations Manager

- Designed a SQL/Python spatial-temporal framework over 500k+ trips to identify cross-dispatch opportunities, surfacing market-specific patterns that improved driver utilization; built incentive-burn pipelines validated against experimental results across US/Canada.

Research Experience

Zuckerman Mind Brain Behavior Institute, Columbia University

Summer 2024 - Present

Research Intern, Computational Neuroscience

Advisor: Prof. Liam Paninski

- Built a semi-supervised pose-estimation pipeline integrating an Ensemble Kalman Smoother with iterative self-training, achieving **10%** improvement over baseline ensembles; manuscript in preparation (pseudo-labeler).

Columbia Irving Medical Center – Mathematical Genomics

Summer 2023

Research Intern

Advisor: Prof. Raúl Rabadán

- Built a hybrid CNN-Transformer for chromatin accessibility prediction over 115 ATAC-seq datasets, reaching Pearson $r = 0.5$ (atac-processing).

Publications

- “Evaluating Chain-of-Thought in Scientific Mediation Problems using Cardiotoxicity Reasoning.” M. Narayan, H Ly. **ICML 2026 Workshop on AI for Science** (under review). Causal-invariance interventions to falsify scientific-reasoning claims of frontier and domain-tuned LLMs (Qwen3-Next-80B, MedGemma-27B).

Open Source

- **FrontierScience benchmark** (Inspect Evals): co-authored the benchmark task implementing OpenAI’s 160-problem expert-reasoning evaluation across physics, chemistry, and biology, with format-specific Olympic and rubric-based Research scorers.
- **Attract-Repel Link Predictor** (PyTorch Geometric, blog post): contributed pseudo-Euclidean attract-repel link prediction for undirected graphs, merged into PyG core.

Skills & Activities

Languages & Tools: Python, SQL, Java, C, Bash; PyTorch, scikit-learn, Lightning AI, FastAPI, dbt, Terraform, Docker, APScheduler; Snowflake, PostgreSQL; AWS, GCP; Git, Linux.

Activities: Ironman Maryland finisher in 14:30 (2.4-mile swim, 112-mile bike, 26.2-mile run).